

UTM TENSILE REPORT — UTM_Test_20260814_120352_100%infill_Videocapture

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A₀) · anchor self-calibration

46.5 MPa

UTS

40.5 MPa

Yield σ_y

2.93 GPa

Modulus E

7.4 %

Failure ϵ_f

2992 kJ/m³

Toughness

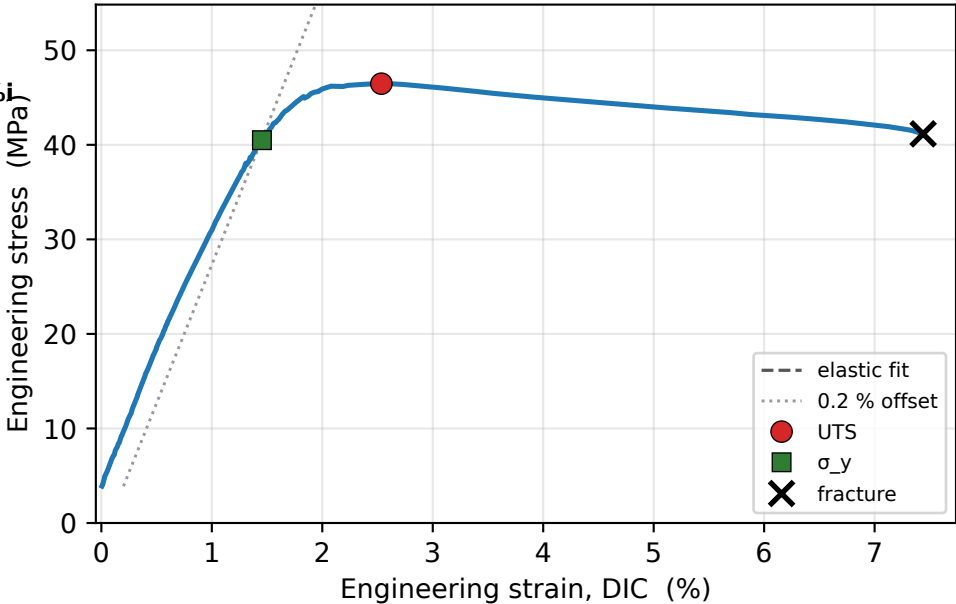
307 N

Preload anchor

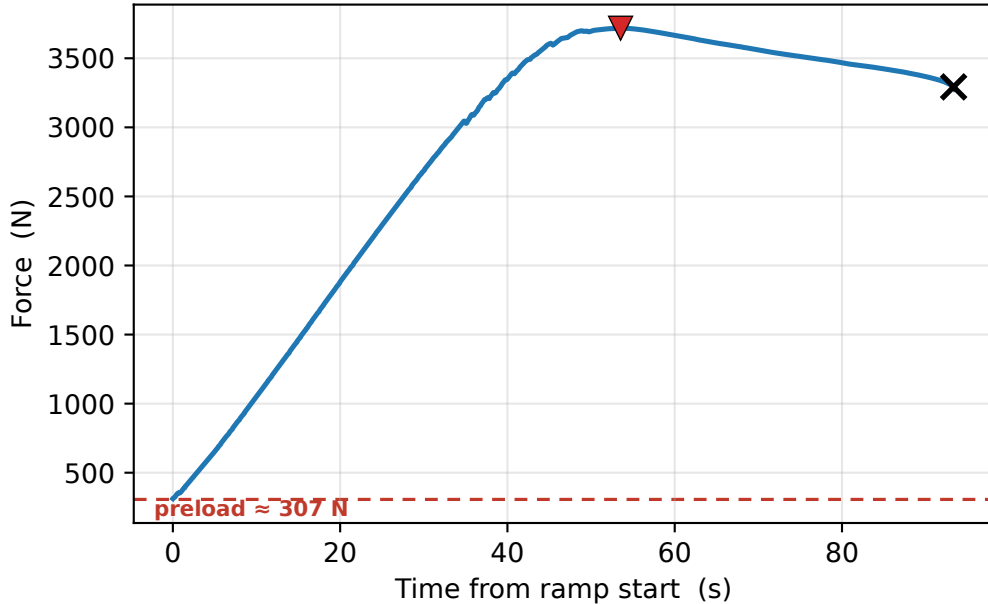
TEST SETTINGS

Specimen	UTM_Test_20260814_120352_100%infill
DIC mode	—
Preload target	—
Test speed	—
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 331.461
Comment	—
Test date	2026-08-14 12:01:18
Duration	188.2 s

Stress-strain



Load vs time

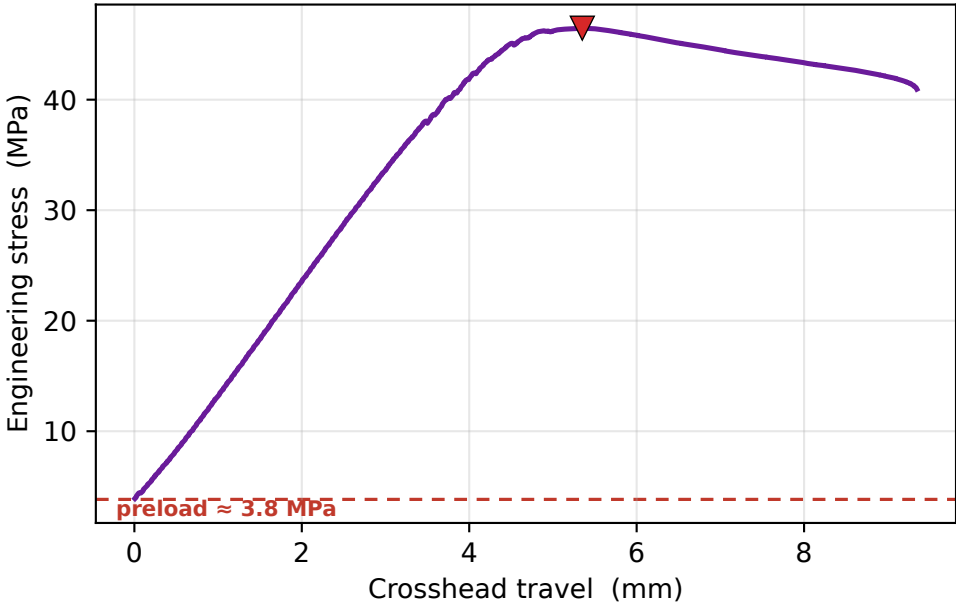


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	46.5	32-60 PASS	×1.25
σ_y	40.5	30-50 PASS	×1.43
E	2.9	3-5.5 low	×0.98
ϵ_f	7.4 %	(range 3-7 %)	×1.08

✓ strength inside journal range (k≈1)

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

